भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 306: Machine Learning Lab
Assignment 1

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1 Problem Domain: State Space Traversal

Consider a system that occupies a state defined by a non-negative integer n. The system transitions between states according to the following rule: If the current state is $k > 0$, the next state is selected via a uniform discrete distribution from the set of all possible preceding states $S = \{0, 1, \ldots, k-1\}$. The system terminates upon reaching the "sink state" 0.

2 Level 1: Trace Analysis

Implement a Python function to simulate the path from state $n = 50$ to state 0. Your script must execute the transition logic 10^4 times and output the arithmetic mean of the path lengths (number of transitions).

- **Focus:** Correct implementation of the state-transition loop and random seed handling.

3 Level 2: Path Parity Determinism

In computational theory, we often need to know the probability of a process satisfying a specific condition. Write an optimized deterministic algorithm to calculate the exact probability that the total path length (number of jumps) is an even number for an initial state $n = 2000$.

- **Constraint 1:** You may not use any probabilistic libraries or simulation.
- **Constraint 2:** The algorithm must maintain a time complexity of $O(n^2)$ or better.

4 Level 3: Computational Scalability

Consider a scenario where the initial state is $n = 10^{500}$. A standard iterative approach or a DP table would exceed the memory and time limits of any physical computer. Write a Python solution that computes the average path length for this n in $O(1)$ time complexity (excluding the overhead of large-integer string parsing).

- **Constraint:** No loops (`for`, `while`) or recursion allowed.

5 Technical Requirements

1. Provide the Python source code for all three levels.
2. Include a brief proof explaining why the state-transition logic leads to a logarithmic growth in path length as $n \to \infty$.
3. Explain how you handled the precision requirements for the $n = 10^{500}$ case without causing a buffer overflow or precision loss.